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Knowledge Interaction Failure in Virtual Communities: Modeling the Impact of Value Co-Destruction on Knowledge System Performance

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Abstract: Virtual communities function as large-scale knowledge interaction systems in which users jointly produce, exchange, and validate knowledge resources. However, not all interactions contribute positively to system performance. This study examined how different forms of value co-destruction behavior degrade knowledge interaction processes and user-level value outcomes in virtual communities. Drawing on survey data from 530 users of firm-hosted virtual communities, a structural equation modeling approach was employed to analyze the effects of five negative interaction behaviors—irresponsible behavior, knowledge hiding, avoidance, conflict, and negative information interaction—on three dimensions of user value: practical, entertainment, and social value. The results indicate that avoidance, conflict, and negative information interaction significantly reduce practical value by impairing knowledge accessibility and information reliability. Knowledge hiding, avoidance, and conflict significantly reduce entertainment and social value by weakening interaction quality and relational embeddedness. Interestingly, irresponsible behavior increases individual entertainment and social value while simultaneously posing systemic risks to collective knowledge quality. These findings suggest that value co-destruction is not merely a behavioral problem but a systemic phenomenon that degrades knowledge flow efficiency, information quality, and collaborative stability in digital knowledge ecosystems. The study contributes to knowledge engineering research by identifying key failure mechanisms in knowledge interaction systems and offers governance implications for designing resilient and sustainable online knowledge platforms.

Keywords: Knowledge interaction systems; Value co-destruction; Knowledge hiding; Information distortion; System degradation

1 Introduction

Brand communities are commercial communities formed around consumers' shared interest in a specific brand, within which brand knowledge and brand-related experiences are exchanged. Such communities have been widely recognized as valuable instruments for marketing, innovation management, and customer relationship management [1]. With the rapid development of the Internet and mobile technologies, virtual brand communities have emerged and transcended geographical constraints, thereby providing favorable conditions for continuous brand knowledge sharing among brand managers, stakeholders, and consumers [2]. From the perspective of brand managers, virtual brand communities enable consumers to participate in product design and development, engage in brand knowledge exchange, and share brand usage experiences. Through these processes, greater knowledge contribution by consumers can be stimulated, allowing communities and firms to obtain richer creative resources and deeper insights into consumer suggestions regarding products and services, ultimately leading to higher returns on community-related investments. From the consumer perspective, high-intensity knowledge interaction within virtual brand communities facilitates access to problem-solving solutions, the exchange of brand-related information and professional knowledge, and the generation of knowledge contributions for oneself, other users, and the brand [3–5]. Accordingly, virtual brand communities function not only as platforms for consumer interaction but also as digital knowledge interaction systems centered on user experience, opinions, and information.

However, knowledge interaction within virtual brand communities does not always fulfill the expectations of community initiators or participating consumers. Divergences in goals and expectations regarding resource allocation,

social norms, and value orientations may give rise to negative interaction behaviors such as knowledge hiding or community avoidance. Moreover, the loosely structured and open nature of virtual communities increases the likelihood of destructive behaviors, including knowledge conflict and negative information interaction. These behaviors may further lead to declines in knowledge quality, structural degradation of the knowledge system, and heightened risks of knowledge conflict and knowledge degeneration. As a consequence, consumers may incur losses of informational, temporal, and status-related resources, resulting in negative emotional responses and diminished perceived value [6, 7]. At the same time, trust in the community and the brand may be weakened, potentially triggering consumer complaints or defection and ultimately undermining both community value and brand value [8, 9]. Interaction processes that lead to the joint reduction or destruction of interaction value have been conceptualized as value co-destruction [10]. From a knowledge engineering perspective, value co-destruction is associated with reduced efficiency of knowledge interaction, increased information conflict, and failures in knowledge collaboration within the communities.

Despite the extensive body of research on value co-creation and its antecedents and consequences, scholarly attention to value co-destruction remains comparatively limited [11]. Even fewer studies have examined how value co-destruction behaviors arising during knowledge interaction processes within virtual brand communities erode consumer value. Existing research has predominantly relied on qualitative approaches, including netnography, the critical incident technique, and case study methods, while quantitative evidence examining the relationships between specific variables and value co-destruction remains scarce [12]. In response to these gaps, firm-hosted virtual brand communities were taken as the empirical context in this study, and the mechanisms through which value co-destruction behaviors emerging from knowledge interaction processes influence consumer value were quantitatively examined. The findings are intended to provide guidance for community managers seeking to mitigate the negative impacts of value co-destruction, foster knowledge interaction among community members, and enhance consumers' knowledge interaction experiences. In doing so, support is offered for the sustainable development of virtual brand communities while simultaneously extending the literature on value co-destruction and consumer value within virtual brand communities conceptualized as digital knowledge interaction systems.

2 Theoretical Background and Research Hypotheses

2.1 Theoretical Background

2.1.1 Concept and dimensions of value co-destruction in virtual brand communities

Plé and Cáceres [13] argued that an implicit assumption of service-dominant logic is that value is jointly created through interaction; logically, such interaction may also lead to the joint destruction of value. Accordingly, the concept of value co-destruction was introduced and defined as an interaction process between service systems that, intentionally or unintentionally, results in a decline in the well-being of at least one system, where service systems may refer to individuals or organizations. From a practice-based perspective, Echeverri and Skalén [10] examined the formation of interaction value and emphasized that value co-creation should not be regarded as the sole outcome of interaction processes. Value co-destruction was therefore conceptualized as the joint destruction or reduction of value arising from interactions between service providers and customers. Vafeas et al. [14] further proposed the concept of value diminution, suggesting that suboptimal value outcomes may be realized as a consequence of resource deficiencies or misuse by one or more interacting actors. This perspective is conceptually aligned with the view advanced by Prior and Marcos-Cuevas [15] within a more complex service ecosystem framework, in which value co-destruction was not treated merely as the opposite of value co-creation but was extended to encompass neutral outcomes and outcomes falling below participants' expectations.

Within virtual brand communities, knowledge-sharing interactions among two or more participants may result in a decline in the well-being of at least one participant. The outcomes of such processes may manifest as negative, neutral, or below-expected states. Accordingly, within the online knowledge interaction system composed of firms, consumers, and virtual brand communities, value co-destruction was defined in this study as behaviors—intentional or unintentional—exhibited by consumers, platform managers, or other consumers during interactions involving the sharing of knowledge, information, or experiences, which lead to a reduction in consumer value or brand value in virtual brand community contexts. From the consumer perspective, a decline in well-being is reflected in reduced perceived value and, in some cases, diminished willingness to engage in community-based knowledge sharing. From the perspective of the virtual brand community as a knowledge interaction system, value co-destruction is manifested through reduced efficiency of brand knowledge interaction, deterioration of knowledge quality, decreased knowledge output, intensified knowledge conflict, and structural damage to the knowledge system. Collectively, these effects weaken the community's capacity for brand knowledge collaboration and ultimately exert a negative influence on brand value.

Scholarly efforts have examined the dimensional structure of customer value co-destruction behaviors. From a dual perspective encompassing both customers and service providers, Guan et al. [16] analyzed value co-destruction behaviors in tourism settings and identified several forms, including dysfunctional interpersonal communication,

negative information interaction, irresponsible customer behavior, employee contract violation, and irresponsible employee behavior. Do and Rahman [17] argued that negative customer engagement behaviors may be categorized into two dimensions: disengaged behavior and negatively engaged behavior. Disengaged behavior was defined as customers' emotional or physical withdrawal from the service process, whereas negatively engaged behavior was characterized as customers' active expression of strong negative experiences toward the service brand or provider, including complaints, negative word-of-mouth, and retaliatory actions. Lv et al. [18] constructed an integrated model of value co-destruction in online tourism communities and proposed six types of customer value co-destruction behaviors: social loafing, knowledge hiding, communication overload, distrust, inappropriate behavior, and conflict.

At present, research specifically addressing customer value co-destruction behaviors in virtual brand community contexts remains limited. Given the absence of highly aligned prior studies, insights from existing research have been drawn upon to clarify the origins and dimensions of customer value co-destruction behaviors [19]. Based on a systematic review of the literature on customer value co-destruction and value co-destruction in virtual brand communities, grounded theory and contextual experience analysis were applied to examine the characteristics of customer value co-destruction behaviors within virtual brand communities, leading to the identification of distinct behavioral dimensions. Five dimensions were delineated: irresponsible customer behavior, knowledge hiding behavior, avoidance behavior, conflict behavior, and negative information interaction behavior. Irresponsible customer behavior refers to unreasonable actions directed toward brands or community staff, such as impoliteness, deliberate provocation, or malicious complaints. Knowledge hiding behavior denotes the intentional concealment or withholding of knowledge requested by others during interaction processes, resulting in insufficient information and knowledge sharing. Avoidance behavior describes customers' gradual withdrawal from the community and disengagement from knowledge-sharing activities. Conflict behavior is characterized by strained interpersonal relationships among customers, including disrespectful conduct or verbal aggression. Negative information interaction behavior refers to the distortion of facts during knowledge or information dissemination, which misleads other customers and damages brand image.

2.1.2 Concept and dimensions of customer value in virtual brand communities

Customer value is defined as customers' subjective trade-off between perceived benefits and costs. It is formed through interactions between customers and the focal object, determined by customers themselves, and influenced by contextual factors and customer experiences [20]. Value co-destruction behaviors exert negative effects on customer value, often triggering adverse emotional responses and leading to reductions in perceived time value and economic value [16], as well as impairments in perceived functional and interpersonal benefits [18]. However, Bowden et al. [21] demonstrated that although negative customer engagement behaviors undermine brand and community value, individual-level perceived value is simultaneously enhanced. In such cases, value co-destruction is manifested as an asymmetric outcome in which value increases for one actor while declining for another. Consequently, the effects of customer value co-destruction behaviors on customer value in virtual brand communities require further examination and empirical validation across specific variable dimensions.

Within virtual brand communities conceptualized as knowledge interaction systems, customer perceived value represents a multidimensional variable and may be regarded as a reflection of customers' evaluations of knowledge interaction effectiveness and interaction environment quality. Drawing on the framework proposed by Xu and Hu [22], customer value in virtual communities was operationalized along three dimensions: practical value, entertainment value, and social value. Practical value refers to the value derived from enhanced cognition or the acquisition of problem-solving solutions through knowledge interaction in virtual brand communities. Entertainment value denotes the value associated with emotional relaxation and enjoyment experienced during knowledge interaction. Social value reflects the value generated through the development of social relationships and the attainment of respect and support within knowledge interaction processes in virtual brand communities.

2.2 Research Hypotheses

2.2.1 Effects of irresponsible customer behavior on customer value

Virtual brand communities provide platforms through which customers may communicate with platform staff to obtain solutions or acquire additional brand-related information. When product or service failures are encountered, unreasonable actions directed toward brands or platform staff—such as verbal aggression or malicious complaints—may be adopted by customers to release negative emotions, thereby reducing emotional strain [23]. At the same time, pressure may be exerted on platform staff in an attempt to obtain remediation or compensation, which may contribute positively to customers' entertainment value and practical value. Dissatisfied community members may also act in a systematic and goal-oriented manner by disseminating negative experiences across multiple platforms or community sections. Through the transmission of such experiences to other users, attention and social support may be sought, while relational networks may be established and mobilized to increase the number of consumers participating in opposition to the brand [24]. Through these processes, customers' social value may be enhanced. From the perspective of knowledge interaction systems, irresponsible customer behavior represents a critical risk factor triggering systemic

value co-destruction. Such behavior may disrupt the norms and atmosphere governing knowledge interaction, resulting in reduced efficiency of knowledge production and problem solving. In addition, irresponsible customer behavior may activate cycles of antagonistic interaction, generating a negative demonstration effect in which confrontational behavior is implicitly rewarded. Furthermore, the negative diffusion associated with irresponsible customer behavior may distort the community's knowledge and information ecology, contaminate the knowledge environment, and ultimately impair the perceived value of other customers. Based on the above reasoning, the following hypotheses were proposed:

H1a: Irresponsible customer behavior has a significant positive effect on practical value.

H1b: Irresponsible customer behavior has a significant positive effect on entertainment value.

H1c: Irresponsible customer behavior has a significant positive effect on social value.

2.2.2 Effects of knowledge hiding behavior on customer value

Virtual communities function as tools through which knowledge is generated, shared, disseminated, and utilized via interpersonal interaction. However, in the absence of formal contractual relationships and effective incentives for knowledge contribution, or when contributors lack confidence in their own knowledge-sharing capabilities, motivation to continue sharing knowledge tends to diminish [25]. Knowledge hiding obstructs other customers' access to information and knowledge, disrupts interpersonal relationships among customers, and fosters cycles of mutual knowledge hiding driven by interpersonal distrust [26]. As a result, knowledge resource networks are weakened, leading to negative effects on customers' practical value and social value. Customers may engage in activities such as sharing information and experiences, posting comments, evaluating products, and refining potential solutions on platforms in order to obtain enjoyment and psychological gratification [27]. However, knowledge hiding behavior impedes the emergence of interactive enjoyment, thereby exerting a negative influence on entertainment value. From the perspective of knowledge interaction systems, knowledge hiding behavior reduces the community's knowledge stock and liquidity, directly undermining the platform's instrumental value in efficiently generating solutions. In addition, customers' willingness to share knowledge is suppressed, resulting in a decline in knowledge productivity at the platform level. Knowledge hiding behavior not only diminishes customers' practical value but also constrains the community's capacity for innovation emergence. In essence, knowledge hiding behavior initiates systemic value co-destruction cycles within the platform: knowledge retention at the individual level is amplified into structural blockages in knowledge flow during community interaction, thereby undermining the core functions of virtual communities as knowledge ecosystems. Based on the above reasoning, the following hypotheses were proposed:

H2a: Knowledge hiding behavior has a significant negative effect on practical value.

H2b: Knowledge hiding behavior has a significant negative effect on entertainment value.

H2c: Knowledge hiding behavior has a significant negative effect on social value.

2.2.3 Effects of avoidance behavior on customer value

Virtual brand communities are characterized by high levels of participation and strong customer–brand reciprocity, through which relationships are typically established by fulfilling customers' emotional and social needs [28]. When excessive negative brand experiences are accumulated, gradual disengagement from the brand community may occur. This process of relationship deterioration is accompanied by psychological, social, and emotional changes experienced by customers [29], thereby exerting negative effects on entertainment value and social value. Moreover, disengagement from virtual brand communities prevents access to brand-related knowledge and information embedded within the community. As a consequence, benefits associated with problem solving and cognitive enrichment are constrained, resulting in adverse effects on practical value. From the perspective of knowledge interaction systems, avoidance behavior should not be interpreted merely as an individual withdrawal decision. Instead, diffusion effects are generated through the community's network structure, whereby avoidance behavior functions as a critical feedback mechanism that triggers and accelerates systemic value decay in virtual brand communities. Avoidance behavior reduces knowledge diversity and interaction density within the community and may also induce latent trust crises and social learning effects. More critically, avoidance behavior disrupts the closed-loop feedback mechanism underlying value co-creation. Based on the above reasoning, the following hypotheses were proposed:

H3a: Avoidance behavior has a significant negative effect on practical value.

H3b: Avoidance behavior has a significant negative effect on entertainment value.

H3c: Avoidance behavior has a significant negative effect on social value.

2.2.4 Effects of conflict behavior on customer value

In virtual brand communities, conflict may initially arise from ideological differences, moral disputes, or competitive institutional arrangements, but may subsequently escalate into emotional or interpersonal conflict. In some cases, such conflict may further evolve into antisocial confrontation aimed at manipulating or provoking individual members or groups into debate, thereby fostering an unpleasant community atmosphere [30]. Conflict behavior not only obstructs normal knowledge and information exchange among customers but also requires additional emotional effort to maintain respect for divergent viewpoints. When such effort is absent, destructive tension may

emerge, triggering anger and aggression and ultimately impairing customers' emotional states [31]. As a result, practical value and entertainment value are negatively affected. At the same time, conflict behavior prevents the establishment and maintenance of interpersonal relationships and undermines perceptions of sincere and friendly communication, thereby exerting a negative influence on perceived interpersonal benefits [18]. From the perspective of knowledge interaction systems, conflict behavior represents one of the most destructive catalysts of value co-destruction within virtual brand community ecosystems. Trust crises induced by conflict behavior lead to a contraction of cooperative willingness, causing the community to degenerate from a "knowledge-sharing pool" into an "information vigilance zone." Under such conditions, the generation of innovative solutions through cross-boundary interaction is severely inhibited, and the community's long-term potential for practical value creation is depleted. Moreover, interactional disharmony resulting from conflict behavior may trigger systemic chain reactions that fundamentally erode the structural stability of the community as a space for knowledge production and relationship cultivation, thereby initiating ecological degradation that is difficult to reverse. Based on the above reasoning, the following hypotheses were proposed:

H4a: Conflict behavior has a significant negative effect on practical value.

H4b: Conflict behavior has a significant negative effect on entertainment value.

H4c: Conflict behavior has a significant negative effect on social value.

2.2.5 Effects of negative information interaction behavior on customer value

In virtual brand communities, exaggerated or false brand-related reviews are primarily produced as a means of releasing negative emotions. Secondary motivations include warning other potential customers, retaliating against firms, and ultimately attracting attention from others [32]. Through negative information interaction behavior, emotional catharsis can be achieved and emotional support from others can be sought, thereby generating entertainment value. In addition, relational ties with other customers may be strengthened through advising or warning others, contributing to the creation of relational value [33]. This exerts a positive effect on customers' entertainment value and social value. However, negative information interaction behavior distorts and falsifies information exchanged among customers, undermines the value of information dissemination, and deteriorates the platform's information ecology [34]. This exerts a negative effect on customers' practical value. As a result, customers' ability to obtain reliable knowledge and effective solutions is impaired, leading to a negative impact on practical value. From a system interaction perspective, negative information interaction behavior not only generates immediately visible information distortion but also fundamentally alters the operational logic of the community ecosystem by activating complex systemic feedback loops. Short-term and localized individual value realization is thereby transformed into long-term and collective public value depletion. For instance, negative information interaction may trigger defensive closed-loop responses by brands and communities, suppressing the system's learning and adaptive capacities and shifting the community from an "open co-creation platform" toward a "brand-controlled information dissemination channel." Under such conditions, brands lose opportunities to access authentic—even critical—user feedback, while customers lose effective channels through which brand decisions may be influenced. Consequently, the system's inherent capacity for collaborative co-evolution is stagnated. Based on the above reasoning, the following hypotheses were proposed:

H5a: Negative information interaction behavior has a significant negative effect on practical value.

H5b: Negative information interaction behavior has a significant positive effect on entertainment value.

H5c: Negative information interaction behavior has a significant positive effect on social value.

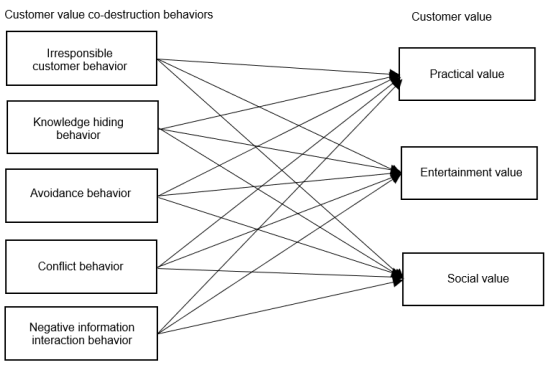

Figure 1. Theoretical framework of the study

Based on the foregoing analysis and the proposed research hypotheses, a conceptual model illustrating the effects of customer value co-destruction behaviors on customer value was developed as the theoretical framework of this study, as shown in Figure 1.

3 Research Design

Table 1. Variables and measurement items

Variable	Sub-Dimension	Item Code	Item Description
Customer value co-destruction behaviors	Irresponsible customer behavior	ICB1	Negative publicity about the brand's products or services is disseminated.
		ICB2	Disrespectful behavior toward community staff is exhibited, such as impatience in tone.
		ICB3	Community staff are mocked as unprofessional and their work performance is disparaged.
	Knowledge hiding behavior	KH1	Others' questions are ignored and responses are not provided.
		KH2	Responses to others' messages are provided, but not in a timely manner.
		KH3	Updated information or posts are deliberately ignored or feigned as unread.
	Avoidance behavior	AB1	The frequency of visiting the community is gradually reduced.
		AB2	Less time is spent in the community.
		AB3	Participation in community activities such as creativity or design is discontinued.
		AB4	Feedback on problems in community management is no longer provided.
	Conflict behavior	CO1	Agreement with other customers on problem solutions or discussion topics is difficult to achieve.
		CO2	Disagreements with other customers escalate into arguments.
		CO3	Content shared by other customers in the community is rejected or viewed negatively.
	Negative information interaction behavior	BIIB1	Inaccurate information is posted within the community.
		BIIB2	Misleading information is posted within the community.
		BIIB3	False information is posted within the community.
		BIIB4	Unverified information is forwarded or disseminated within the community.
Customer value	Practical value	PV1	Required information or knowledge is obtained from the community website or other members.
		PV2	Usage experiences and insights shared by other members are perceived as highly beneficial.
		PV3	Problems encountered are resolved through posting, private messaging, or seeking help within the community.
	Entertainment value	EV1	Participation in the community helps pass otherwise boring time.
		EV2	Participation in the community contributes to emotional relaxation.
		EV3	Participation in the community provides relief from stress and responsibilities.
	Social value	SV1	New friends are made through the community.
		SV2	A sense of achievement is gained through the community.
		SV3	Self-image is enhanced through participation in the community.
		SV4	Respect or appreciation from others is obtained through the community.

3.1 Variable Measurement

To enhance the reliability and validity of the questionnaire, established and widely used measurement scales from virtual brand community research were adopted for customer value variables. Measurement items for customer value co-destruction behaviors in virtual brand communities were developed with reference to validated scales reported in prior studies. The questionnaire was structured into two main sections. The first section collected respondents' basic information, including gender, age, educational level, length of membership in the virtual brand community, average weekly frequency of community use, and average duration of each visit. The second section focused on the measurement of the hypothesized variables. Specifically, customer value co-destruction behaviors in virtual brand communities were measured using 17 items adapted from prior research, while customer value was measured using 10 items designed primarily based on the scale developed by Xu and Hu [22]. Except for the basic information items, all questionnaire items were measured using a five-point Likert scale. The detailed constructs and item descriptions are presented in Table 1.

3.2 Data Collection

A questionnaire survey method was employed in this study, and users with prior experiences of value co-destruction in virtual brand communities were invited to participate. To ensure the rationality and effectiveness of the questionnaire, a pilot study was conducted in advance. Based on the pilot results, revisions were made to item content and wording accuracy, and the reliability and validity of the measurement items were preliminarily assessed. During the formal survey phase, data were collected using the Wenjuanxing sample service, resulting in 530 valid questionnaires. In the final sample, 40.8% of respondents were male and 59.2% were female. With respect to age distribution, 0.8% were under 20 years old, 68.1% were between 21 and 30 years old, 27.7% were between 31 and 40 years old, and 3.4% were over 40 years old. In terms of educational attainment, 0.6% had completed high school or below, 5.1% held a junior college degree, 87.5% held a bachelor's degree, and 6.8% held a master's degree or above. Regarding the duration of membership in virtual brand communities, 9.2% of respondents had been members for one year or less, 48.9% for two to three years, 28.1% for three to five years, and 13.8% for more than five years. In terms of average weekly usage frequency, 30.8% reported using the community fewer than three times per week, 55.8% reported four to seven times per week, and 13.4% reported more than seven times per week. With respect to average time spent per visit, 25.1% reported less than 30 minutes, 64.5% reported between 30 minutes and one hour, and 10.4% reported more than one hour.

4 Data Analysis and Results Discussion

4.1 Reliability and Validity Assessment

Confirmatory factor analysis (CFA) was conducted using AMOS 24.0 to assess the reliability and validity of the measurement items for customer value co-destruction behaviors and customer value in virtual brand communities. The goodness-of-fit indices for the measurement model of customer value co-destruction behaviors were as follows: the normed chi-square (NC,χ^2/df) = 2.002, p = 0.000, root mean square error of approximation (RMSEA) = 0.044, goodness-of-fit index (GFI) = 0.953, adjusted goodness-of-fit index (AGFI) = 0.934, normed fit index (NFI) = 0.944, comparative fit index (CFI) = 0.971, and incremental fit index (IFI) = 0.971. These results indicate an acceptable fit between the proposed model and the observed data, suggesting that the model adequately fits and predicts the underlying quantitative relationships among the constructs. The CFA results for customer value co-destruction behaviors in virtual brand communities are presented in Table 2. The composite reliability (CR) values and Cronbach's α coefficients for the five dimensions of customer value co-destruction behaviors all exceeded 0.75, indicating satisfactory internal consistency reliability. In addition, the standardized factor loadings for all measurement items were greater than 0.60, and the average variance extracted (AVE) values for all constructs exceeded the threshold of 0.50, demonstrating adequate convergent validity. For each construct, the square root of the AVE was greater than the corresponding inter-construct correlation coefficients, supporting discriminant validity.

The goodness-of-fit indices for the customer value measurement model in virtual brand communities were as follows: NC (χ^2/df) = 1.955, p = 0.001, RMSEA = 0.051, GFI = 0.978, AGFI = 0.962, NFI = 0.974, CFI = 0.987, and IFI = 0.987. These indices indicate a satisfactory fit between the measurement model and the observed data, suggesting that the model is capable of adequately fitting and predicting the underlying quantitative relationships among the constructs. The CFA results for customer value in virtual brand communities are presented in Table 3. The CR values and Cronbach's α coefficients for all three dimensions of customer value exceeded 0.75, demonstrating satisfactory internal consistency reliability. In addition, all standardized factor loadings were greater than 0.70, and the AVE values for all constructs exceeded the recommended threshold of 0.50, thereby supporting convergent validity. Discriminant validity was further confirmed, as the correlation coefficients between each construct and the remaining two constructs were smaller than the square root of the corresponding AVE values.

Table 2. Confirmatory factor analysis (CFA) results for customer value co-destruction behaviors in virtual brand communities

Variable	Item	Std. loading	t-value	CR	AVE	Cronbach's α	Correlation coefficients				
							ICB	KH	AB	CO	BIIB
Irresponsible customer behavior (ICB)	ICB1	0.745	14.692***	0.777	0.537	0.776	0.733				
	ICB2	0.701	14.411***	0.777	0.537	0.776	0.733				
	ICB3	0.752	—	0.777	0.537	0.776	0.733				
Knowledge hiding (KH)	KH1	0.816	15.740***	0.781	0.546	0.775	0.621	0.739			
	KH2	0.633	12.797***	0.781	0.546	0.775	0.621	0.739			
	KH3	0.756	—	0.781	0.546	0.775	0.621	0.739			
Avoidance behavior (AB)	AB1	0.834	14.012***	0.846	0.582	0.842	0.508	0.609	0.763		
	AB2	0.849	14.224***	0.846	0.582	0.842	0.508	0.609	0.763		
	AB3	0.738	13.444***	0.846	0.582	0.842	0.508	0.609	0.763		
	AB4	0.607	—	0.846	0.582	0.842	0.508	0.609	0.763		
Conflict behavior (CO)	CO1	0.687	14.240***	0.788	0.554	0.780	0.573	0.569	0.458	0.744	
	CO2	0.718	15.508***	0.788	0.554	0.780	0.573	0.569	0.458	0.744	
	CO3	0.822	—	0.788	0.554	0.780	0.573	0.569	0.458	0.744	
Negative information interaction behavior (BIIB)	BIIB1	0.701	15.301***	0.838	0.565	0.833	0.513	0.431	0.220	0.485	0.752
	BIIB2	0.746	15.867***	0.838	0.565	0.833	0.513	0.431	0.220	0.485	0.752
	BIIB3	0.796	17.211***	0.838	0.565	0.833	0.513	0.431	0.220	0.485	0.752
	BIIB4	0.760	—	0.838	0.565	0.833	0.513	0.431	0.220	0.485	0.752

Note: Values on the diagonal represent the square root of the AVE for each construct; *** indicates $p \leq 0.001$. CR: composite reliability; AVE: average variance extracted

Table 3. Confirmatory factor analysis (CFA) results for customer value in virtual brand communities

Variable	Item	Std. Loading	t-Value	CR	AVE	Cronbach's α	Correlation Coefficients		
							PV	EV	SV
Practical value (PV)	PV1	0.740	14.555***	0.797	0.567	0.796	0.753		
	PV2	0.803	14.499***	0.797	0.567	0.796	0.753		
	PV3	0.714	—	0.797	0.567	0.796	0.753		
Entertainment value (EV)	EV1	0.751	16.287***	0.808	0.584	0.807	0.630	0.764	
	EV2	0.769	16.015***	0.808	0.584	0.807	0.630	0.764	
	EV3	0.773	—	0.808	0.584	0.807	0.630	0.764	
Social value (SV)	SV1	0.720	17.447***	0.860	0.607	0.859	0.585	0.689	0.779
	SV2	0.792	19.732***	0.860	0.607	0.859	0.585	0.689	0.779
	SV3	0.774	19.150***	0.860	0.607	0.859	0.585	0.689	0.779
	SV4	0.827	—	0.860	0.607	0.859	0.585	0.689	0.779

Note: Values on the diagonal represent the square root of the AVE for each construct; *** indicates $p \leq 0.001$. CR: composite reliability; AVE: average variance extracted

4.2 Path Analysis and Hypothesis Testing

Structural equation modeling was conducted using AMOS 24.0 to test the hypothesized effects of customer value co-destruction behaviors on customer value. The results are reported in Table 4. The standardized path coefficient from irresponsible customer behavior to practical value was small (p = 0.729), indicating that although a positive association was estimated, the effect was not supported empirically. This outcome may be explained by firms' heterogeneous response strategies to irresponsible customer behavior. In practice, conciliatory approaches—such as emotional appeasement and assurances of non-recurrence—may be adopted, while defensive strategies—such as refusing unreasonable requests—may also be implemented [35]. Under such conditions, remediation or compensation may not be obtained through irresponsible behavior. The standardized path coefficient from knowledge hiding behavior to practical value was also small (p = 0.189). Although a negative association was estimated, empirical support was not observed. This result may be attributed to the fact that participation in virtual spaces may occur in the form of pure knowledge consumption, whereby individuals browse content without engaging in knowledge

sharing [25]. Such "free-riding" behavior does not necessarily hinder access to information within virtual brand communities and therefore may not significantly reduce practical value. With respect to negative information interaction behavior, the standardized path coefficients to entertainment value (p = 0.987) and social value (p = 0.362) were both small. Although positive effects were estimated, the hypothesized relationships were not supported. One plausible explanation is that distorted or misleading information may be quickly identified and publicly challenged by other community members, thereby preventing these behaviors from achieving the expected outcomes intended by the consumers.

Accordingly, the model was further refined by removing statistically insignificant paths. The standardized path coefficients and hypothesis testing results of the revised model are presented in Table 4. As shown, hypotheses H1b, H1c, H2b, H2c, H3a, H3b, H3c, H4a, H4b, H4c, and H5a were supported. Specifically, irresponsible customer behavior exerted significant positive effects on entertainment value and social value; knowledge hiding behavior exerted significant negative effects on entertainment value and social value; avoidance behavior exerted significant negative effects on practical value, entertainment value, and social value; conflict behavior exerted significant negative effects on practical value, entertainment value, and social value; and negative information interaction behavior exerted a significant negative effect on practical value.

Based on the results of the structural equation path analysis, the theoretical framework was refined, as illustrated in Figure 2.

Table 4. Hypothesis testing results for the effects of customer value co-destruction behaviors on customer value in virtual brand communities

Path	Before removing insignificant paths			After removing insignificant paths		
	Coeff.	**t-value**	**Result**	**Coeff.**	**t-value**	**Result**
H1a: Irresponsible customer behavior → Practical value	0.032	0.346	Not supported	—	—	—
H1b: Irresponsible customer behavior → Entertainment value	0.176	2.058^{*}	Supported	0.161	2.134^{*}	Supported
H1c: Irresponsible customer behavior → Social value	0.241	2.659^{**}	Supported	0.243	3.268^{**}	Supported
H2a: Knowledge hiding behavior → Practical value	−0.127	−1.314	Not supported	—	—	—
H2b: Knowledge hiding behavior → Entertainment value	−0.327	-3.666^{***}	Supported	−0.290	-3.579^{***}	Supported
H2c: Knowledge hiding behavior → Social value	−0.299	-3.206^{**}	Supported	−0.232	-2.962^{**}	Supported
H3a: Avoidance behavior → Practical value	−0.155	-2.056^{*}	Supported	−0.207	-3.402^{***}	Supported
H3b: Avoidance behavior → Entertainment value	−0.313	-4.300^{***}	Supported	−0.329	-4.747^{***}	Supported
H3c: Avoidance behavior → Social value	−0.376	-5.103^{***}	Supported	−0.413	-5.924^{***}	Supported
H4a: Conflict behavior → Practical value	−0.208	-2.482^{*}	Supported	−0.223	-2.958^{**}	Supported
H4b: Conflict behavior → Entertainment value	−0.182	-2.524^{*}	Supported	−0.188	-2.745^{**}	Supported
H4c: Conflict behavior → Social value	−0.167	-2.198^{*}	Supported	−0.161	-2.296^{*}	Supported
H5a: Negative information interaction behavior → Practical value	−0.158	-2.264^{*}	Supported	−0.193	-3.318^{***}	Supported
H5b: Negative information interaction behavior → Entertainment value	−0.001	0.987	Not supported	—	—	—
H5c: Negative information interaction behavior → Social value	0.057	0.911	Not supported	—	—	—
Model fit	NC (χ^2/df) = 2.136, $p = 0.000$, RMSEA = 0.046, GFI = 0.918, AGFI = 0.896, NFI = 0.906, CFI = 0.947, and IFI = 0.948			NC (χ^2/df) = 2.116, $p = 0.000$, RMSEA = 0.046, GFI = 0.918, AGFI = 0.897, NFI = 0.906, CFI = 0.948, and IFI = 0.948		

Note: * indicates $p \leq 0.05$, ** indicates $p \leq 0.01$, and *** indicates $p \leq 0.001$.

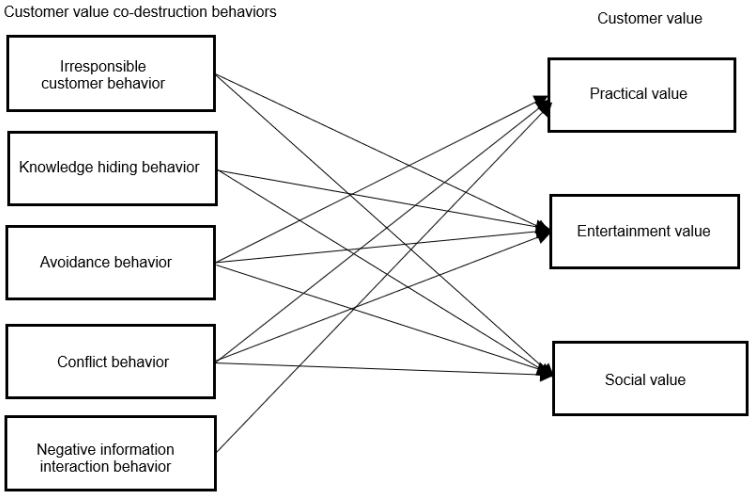

Figure 2. Revised theoretical framework

5 Conclusions and Prospects

5.1 Conclusions

Within the context of virtual brand communities, a structural equation modeling approach was employed to construct the impact pathways linking customer value co-destruction behaviors to customer value. The principal conclusions are summarized below. First, the directions of influence exerted by different dimensions of customer value co-destruction behaviors on customer value were found to vary. Avoidance behavior, conflict behavior, and negative information interaction behavior were shown to have significant negative effects on practical value. Knowledge hiding behavior, avoidance behavior, and conflict behavior exerted significant negative effects on entertainment value, whereas irresponsible customer behavior exerted a significant positive effect on entertainment value. Similarly, knowledge hiding behavior, avoidance behavior, and conflict behavior were found to significantly reduce social value, while irresponsible customer behavior exerted a significant positive effect on social value.

These findings are not fully consistent with theoretical arguments suggesting that value co-destruction necessarily leads to a decline in well-being [13], nor do they fully align with evidence indicating that negative customer engagement may enhance individual perceived value [21]. Instead, the results indicate that, when the effects of value co-destruction are examined across multiple dimensions, analysis should be conducted at the level of specific value dimensions rather than at an aggregate level. Second, divergent effects were observed across different value dimensions for the same type of customer value co-destruction behavior. Negative information interaction behavior exerted a significant negative effect on practical value, whereas its effects on entertainment value and social value were not statistically significant but exhibited opposite directional tendencies. This finding provides an empirical response to the question raised by Plé [36] regarding whether a single behavior may simultaneously generate one form of value while destroying another.

5.2 Theoretical Contributions and Managerial Implications

The theoretical contributions of this study are reflected in three main aspects. First, the effects of customer value co-destruction behaviors on customer value were examined within the context of virtual brand communities conceptualized as knowledge interaction systems. From an empirical perspective, negative knowledge interaction behaviors in digital knowledge communities were revealed, thereby providing a theoretical basis for understanding interaction failure mechanisms in knowledge interaction systems. Second, it was empirically demonstrated that knowledge interaction behaviors in virtual brand communities may not only co-create customer value but may also co-destroy it. This finding provides theoretical support for understanding how the erosion of customer value may be avoided during knowledge interaction processes in virtual brand community contexts. Third, the results concerning customer value co-destruction behaviors in virtual brand communities indicate that negative information interaction behavior reduces practical value while simultaneously enhancing entertainment value and social value. This pattern empirically confirms that a single behavior may lead to the creation of one form of value while simultaneously destroying another.

From a knowledge interaction system governance perspective, the managerial implications for platform administrators are manifested in several key aspects as follows:

First, irresponsible customer behavior may be interpreted as a typical manifestation of the "tragedy of the commons" within virtual communities. Actions undertaken by individual customers in pursuit of short-term private benefits

undermine the public resources that sustain long-term community prosperity—such as trust, rational interaction climates, brand investment, and high-quality engagement—ultimately resulting in collective value loss. Platform governance is therefore required to strike a balance between respecting users' rights to expression and preserving a healthy community ecosystem. This balance may be achieved through clearly articulated community norms, impartial mediation mechanisms, active responses to constructive feedback, and effective identification and restraint of malicious behaviors, thereby guiding interactions toward value co-creation rather than value co-destruction.

Second, knowledge hiding behavior should be regarded not merely as an issue of individual motivation, but as a critical systemic risk affecting the sustainable development of virtual communities. To mitigate such risk, platform administrators are required to design appropriate incentive mechanisms, establish trust-building infrastructures, cultivate a culture of sharing, and employ algorithmic tools to enhance knowledge visibility and recognize valuable contributions. Through these measures, cycles of value co-destruction can be disrupted and redirected toward a virtuous ecosystem of value co-creation.

Third, avoidance behavior functions as an early warning signal of the health of the virtual brand community ecosystem. It reveals structural deficiencies in the system's ability to maintain trust, address negative experiences, and transform critical feedback into drivers of innovation. Rather than focusing solely on "preventing churn," platform governance should be oriented toward building more inclusive, responsive, and resilient interaction mechanisms. Such mechanisms may include the establishment of effective channels for processing negative experiences, the design of procedures that enable critical opinions to be expressed constructively, and the creation of flexible participation spaces that allow users to disengage temporarily and re-enter the community with minimal barriers. In this way, tendencies toward avoidance may be transformed into opportunities for system iterative upgrading, thereby sustaining the long-term vitality of the community as a value co-creation ecosystem.

Fourth, conflict management represents a core challenge in the governance of virtual brand communities. Effective governance requires the capability to distinguish between constructive debate and destructive conflict, as well as the establishment of rapid and effective intervention mechanisms. These mechanisms include the formulation of clear and widely communicated baseline rules for interaction, the training of community managers in neutral mediation and emotional regulation, and the design of platform features (e.g., consensus-building tools and topic segmentation) to reduce the visibility and contagion of unproductive conflict. More importantly, a problem-solving–oriented collaborative culture should be actively cultivated and rewarded so that potential disagreements may be redirected toward systematic value co-creation processes that deepen knowledge and improve products, thereby safeguarding the long-term health and prosperity of the community ecosystem.

Fifth, governance of negative information interaction behavior should not be confined to the technical level of content moderation, but should instead be addressed at the level of systemic interaction dynamics. This requires, first, the construction of balanced mechanisms that integrate incentives for authenticity with norms for emotional expression—for example, by enhancing the visibility of reliable information through identity verification, purchase-history linkage, or weighted voting, while guiding emotional expression within constructive frameworks. In addition, transparent systems for information traceability and correction should be established, enabling community members to flag questionable information, engage in discussion, and contribute factual supplementation, thereby transforming the truth-seeking process itself into a valuable form of public deliberation. Furthermore, a form of "controlled openness" between brands and communities should be maintained, such that authentic user feedback—including sharp criticism—remains accessible, while excessive defensiveness that severs the lifeline of value co-creation is avoided. Only through such systemic governance can the destructive potential of negative information interaction be redirected toward the development of a more resilient, trustworthy, and learning-oriented community information ecosystem.

5.3 Research Limitations and Prospects

Several limitations are associated with this study, which should be addressed in future research. First, moderating and mediating effects were not incorporated into the analytical framework. Future studies may introduce variables such as customer demographic characteristics, community identification, and sense of community belonging into the influence pathways. Second, the empirical analysis was conducted within the context of firm-hosted virtual brand communities. Whether the conclusions remain applicable to transactional communities or communities formed spontaneously by members warrants further empirical examination.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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